LECTURE 1.5 · SHIP AND OCEAN SYSTEMS DESIGN

# The Design Equation

Papanikolaou (2026) §2.13, Appendix C · Session W1.4 · Lecture 20 min + Exercise 1.5 60 min

**Luís Batista**

Instituto Superior Técnico / University of Lisbon

luis@veranavis.com

calendly.com/vnsd-luisbatista/meetme

# One Equation, Two Sides

### Buoyancy: the hull

$$\Delta = \rho\, C_B\, L\, B\, T\, k_A$$

kA corrects the moulded volume for shell plating and appendages.

### Weight: the ship and its load

$$\Delta = \mathrm{DWT} + W_{ST} + W_{OT} + W_M$$

Steel, outfit and machinery all grow with the ship, so Δ sits on both sides.

In Exercise 1.4, Δ came from the deadweight and L, B, T from separate regressions. Put those L, B and T in the buoyancy side and you do not get back the same Δ: the two sides **miss by a few percent**. Today we choose the size so that both sides give the same Δ: that is the design equation.

# Fix the Ratios, and One Dimension Is Left

## Deadweight carriers: from Δ

Take L/B and B/T from similar ships and write L and T through B:

$$L = \left(\frac{L}{B}\right) B, \qquad T = \frac{B}{(B/T)}$$

Buoyancy then has one unknown, B:

$$\Delta = \rho\, C_B\, k_A\, \frac{(L/B)}{(B/T)}\, B^3$$

$$B = \left[\frac{\Delta\,(B/T)}{\rho\, C_B\,(L/B)\, k_A}\right]^{1/3} \qquad L = \left[\frac{\Delta\,(L/B)\,(L/T)}{\rho\, C_B\, k_A}\right]^{1/3}$$

Then T = B/(B/T) and D = B/(B/D).

## Volume carriers: from the hull volume

The same steps, with B/D in place of B/T:

$$B = \left[\frac{\nabla_D\,(B/D)}{C_{BD}\,(L/B)}\right]^{1/3}$$

**Note · the hull volume to the deck**

$$\nabla_D = C_{BD}\; L\, B\, D$$

Moulded volume up to the main deck, at depth D: not the displaced volume, which stops at the draught T.

$$C_{BD} = C_B + C_1\,\frac{D - T}{T}\,(1 - C_B)$$

$C_1 \approx 0.25$ for little flare above the waterline, 0.4 to 0.7 for strong flare.

# The Reference Ship: Closing the Gap

DWT 37,600 t, 14.5 kn, draught limit 11.5 m (0.6, slide 2) · tanker fits of Appendix A.4 · ρ = 1.025 t/m³, kA = 1

**Given by Exercise 1.4 (solved in class)**

Δ = 46,050 t (Fig. A.51) · L 174.7 · B 29.50 · T 10.97 · D 14.95 m (Figs. A.53–A.56)

L/B 5.92 · B/T 2.69 · B/D 1.97 · CB 0.757 (Horn, Fn 0.180)

| | | |
|---|---|---|
| 1 | The gap: buoyancy with those values | $\rho C_B L B T = 1.025 \times 0.757 \times 174.7 \times 29.50 \times 10.97 = 43{,}880\ \text{t}\ \ (-4.7\,\%)$ |
| 2 | Solve the design equation for B | $B = \left[\dfrac{46{,}050 \times 2.69}{1.025 \times 0.757 \times 5.92}\right]^{1/3} = 30.0\ \text{m}$ |
| 3 | L, T and D from the ratios | $L = 5.92 \times 30.0 = 177.6\ \text{m},\quad T = \dfrac{30.0}{2.69} = 11.15\ \text{m},\quad D = \dfrac{30.0}{1.97} = 15.2\ \text{m}$ |
| 4 | Check: both sides and the brief | $\rho C_B L B T = 46{,}050\ \text{t} = \Delta,\qquad T = 11.15\ \text{m} < 11.5\ \text{m}$ |

# Normand: Each Weight Group Scales with Δ

| Weight group | Grows with | Power of Δ |
|---|---|---|
| Steel structure, outfitting | the hull | 1 |
| Machinery | $P \propto \Delta^{2/3} \cdot V^3$ | 2/3 |
| Fuel | $P \cdot R/V$ | 2/3 |
| Crew and effects | installed power | 2/3 |
| Payload, passengers, stores | the owner | 0 |

$$\Delta = A\,\Delta + B\,\Delta^{2/3} + C$$

A, B and C come from a parent ship whose weight groups are known.

Valid for deviations from the parent of up to **10%**, exceptionally 20%. Beyond that, exponents by ship type: steel 0.92 for tankers, 0.98 for bulk carriers.

# Solving It: Iterate Until the Ship Floats

**1**

**Write it as**

$$\Delta = g(\Delta)$$

$$g(\Delta) = \mathrm{DWT} + W_L(\Delta)$$

The weight of a ship of size Δ. Start at $\Delta_0$ = DWT, a ship with no lightship.

**2**

**Iterate**

$$\Delta_{k+1} = g(\Delta_k)$$

Add up the weights of the current ship; that total is the displacement of the next one.

**3**

**Check**

$$|g'(\Delta)| < 1$$

Each extra tonne of ship must cost less than a tonne of its own lightship, or the loop never closes.

$$g'(\Delta) = \frac{1}{\Delta} \sum_i n_i \, W_i$$

$n_i$ is the exponent of weight group $W_i$. With exponents of 1 or less and a lightship smaller than Δ, g′ stays below 1: the iteration converges. Newton–Raphson converges faster: homework H1.

EXERCISE 1.5 · INDIVIDUAL · 60 MIN · PRACTICE FOR D3

# Size the Reference Tanker by Hand

**The reference ship** (brief: 0.6) · DWT 37,600 t · V 14.5 kn · from Exercise 1.4: CB 0.757 · L/B 5.92 · B/T 2.69 · D/T 1.36 · W_M in t with P in kW

$$W_{ST} = 0.032\,E^{1.36} \qquad E = L\,(B+T) + 0.85\,L\,(D-T) \qquad W_{OT} = 0.025\,\Delta \qquad W_M = 0.10\,P \qquad P = \frac{\Delta^{2/3}\,V^3}{450}$$

### 1 Equilibrium · 5 min

Known and unknown; why $\Delta_0$ = DWT is a lower bound

### 2 Dimensions · 5 min

L, B, T and D as functions of Δ

### 3 Steel · 7 min

$E = c_E\,\Delta^{2/3}, \quad W_{ST} = c_{st}\,\Delta^n$

Find the two coefficients and the exponent n

### 4 Iteration · 6 min

g(Δ) with every coefficient evaluated

### 5 Convergence · 10 min

Prove it converges for any Δ ≥ DWT, without iterating

### 6 Solution · 15 min

Iterate to a change below 1 t: Δ*, L×B×T×D, weight groups

**Check (7 min):** the worked answer is shown at the end (Exercise Answers – Week 1). Not graded. Q7–8: homework H1.

**Your project:** your group applies this to its own ship outside class and completes D3 (see your Project Scope Sheet).

# Three Things to Take Away

**The design equation** turns buoyancy = weight into one dimension, once the ratios and CB come from similar ships.

**Normand** writes every weight group as a power of Δ, from a parent ship within 10%:

$$\Delta = A\,\Delta + B\,\Delta^{2/3} + C$$

**Iterate** from $\Delta_0$ = DWT; it converges because the lightship grows more slowly than the ship.

**Next:** Wed 11 Nov, human factors in ship design (guest lecturer). Fri 13 Nov (W2.3): length, beam, depth, draught and form coefficients. Read §2.3–2.12.